



4.5.2 Rank Revealing Householder QR factorization

The unblocked QR factorization discussed in [Section 3.3](#) can be supplemented with column pivoting, yielding `HQRP_unb_var1` in [Figure 4.5.2.1](#). In that algorithm, we incorporate the idea that the weights that are used to determine how to pivot can be updated at each step by using information in the partial row r_{12}^T , which overwrites a_{12}^T , just like it was in [Subsection 4.5.1](#).

```
[A, t, p] = HQRP_unb_var1(A)
v := ComputeWeights(A)
A → ( ATL | ATR
      ABL | ABR ), t → ( tT
                        tB ), p → ( pT
                                pB ), v → ( vT
                                           vB )
ATL is 0 × 0 and tT has 0 elements
while n(ATL) < n(A)
    ( ATL | ATR
      ABL | ABR ) → ( A00 | a01 A02
                      a10T | α11 a12T
                      A20 | a21 A22 ), ( tT
                                         tB ) → ( t0
                                                  t1
                                                  t2 ),
    ( pT
      pB ) → ( p0
               π1
               p2 ), ( vT
                     vB ) → ( v0
                              ν1
                              v2 )
    [ ( ν1
        v2 ), π1 ] = DeterminePivot( ( ν1
                                         v2 ) )
    ( a01 A02
      α11 a12T
      a21 A22 ) := ( a01 A02
                      α11 a12T
                      a21 A22 ) P(π1)T
    [ ( α11
        a21 ), τ1 ] := [ ( ρ11
                           u21 ), τ1 ] = Housev ( α11
                                                    a21 )
    w12T := (a12T + a21HA22)/τ1
    ( a12T
      A22 ) := ( a12T - w12T
                 A22 - a21w12T )
    v2 = UpdateWeight(v2, a12)
    ...
endwhile
```

Figure 4.5.2.1. Rank Revealing Householder QR factorization algorithm.

Combining a blocked Householder QR factorization algorithm, as discussed in [Subsubsection 3.4.1.3](#), with column pivoting is tricky, since half the computational cost is inherently in computing the parts of R that are needed to update the weights and that stands in the way of a true blocked algorithm (that casts most computation in terms of matrix-matrix multiplication). The following papers are related to this:

- [34] Gregorio Quintana-Orti, Xiaoabai Sun, and Christof H. Bischof, A BLAS-3 version of the QR factorization with column pivoting, SIAM Journal on Scientific Computing, 19, 1998.

discusses how to cast approximately half the computation in terms of matrix-matrix multiplication.

- [26] Per-Gunnar Martinsson, Gregorio Quintana-Orti, Nathan Heavner, Robert van de Geijn, Householder QR Factorization With Randomization for Column Pivoting (HQRRP), SIAM Journal on Scientific Computing, Vol. 39, Issue 2, 2017.

shows how a randomized algorithm can be used to cast most computation in terms of matrix-matrix multiplication.